

SmartCare v0.5 - Architecture and Refactoring Workbook

Week 8 student resource

Current Architecture Problems

Problem	Evidence	Impact	Refactoring
User interface is mixed with database	input() and SQL INSERT are in the same function	We can't test the code without a person typing on the keyboard.	Move the SQL code to the Persistence layer.
Rules in the menu	Conflict checks happen in the UI file	The code is messy and hard to reuse.	Move the business rules to the Domain layer.
Direct database link	import sqlite3 is inside the Service	The system is permanently locked to SQLite.	Create a Repository interface to hide the DB.

Layer Responsibilities

Layer	Responsibilities	Must not contain
Presentation	Gets user input and prints text.	SQL queries or business rules.
Service	Runs the main step-by-step workflows.	input(), print(), or SQL.
Domain	Holds core rules and basic classes.	Database connections or UI code.
Repository	Acts as a bridge for getting data.	UI code or main workflow logic.
Persistence	Does the actual SQL saving.	Business rules or text formatting.

Architecture Diagram

Insert SmartCare v0.5 architecture and dependency direction.

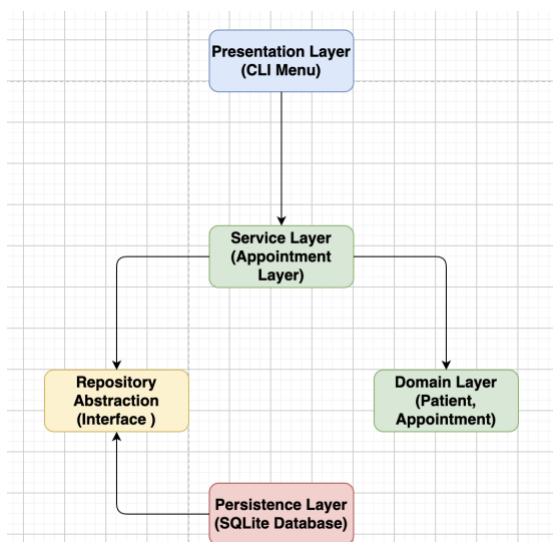


Figure 1 (created using draw.io)

SOLID Review

Principle	Relevant?	Evidence	Decision
SRP	Yes	The old main file did everything.	Accept. We are splitting it into layers.
OCP	No	We aren't adding new types of things yet.	Defer until we need to add new features.
LSP	No	We aren't using inheritance.	Not needed for this project.
ISP	Yes	We only need a save and find method.	Accept. We will keep the repository interface small.
DIP	Yes	The service relied directly on SQLite.	Accept. The service will use the repository instead.

AI Architecture Review

AI suggestion	Observed problem?	Decision	Reason	Verification
Repository Pattern	Yes, SQL was stuck in the Service	Accepted	It hides the database so the logic is cleaner.	The Service runs without importing SQLite.
Microservices	No	Rejected	It makes the system too big and hard to maintain.	Kept the code in one simple project.
Big CRUD Interface	No	Deferred	Kept “save” and “find”, but removed the rest.	We don't need update/delete yet.